

Databases 1

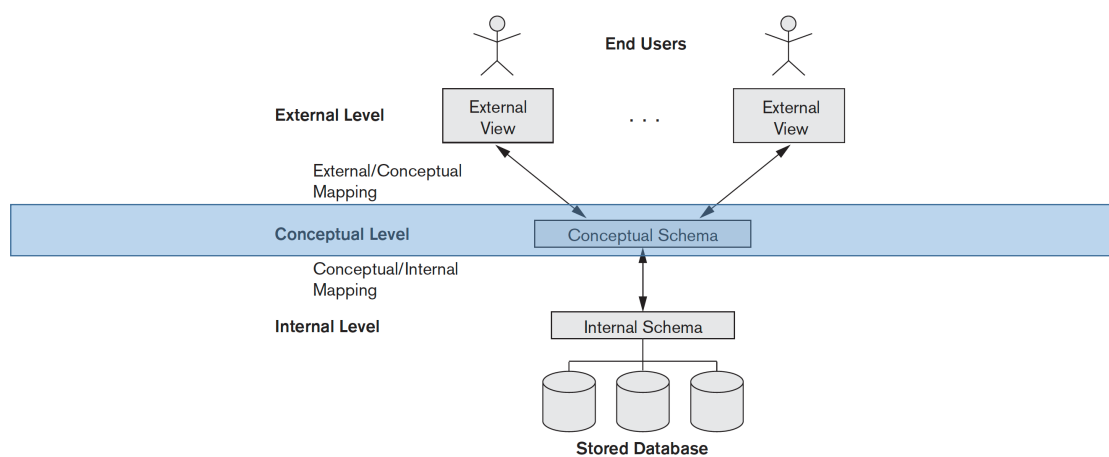
dr hab. inż. Przemysław Korytkowski, prof. ZUT

Lecture 2 – Conceptual Data Modeling and Database Design, ER Diagrams

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Three schemas architecture



source: Elmasri, R., & Navathe, S. (2014). *Fundamentals of database systems* (Vol. 7). Pearson.

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conceptual data modeling = database design

- Refers to the practices and strategies for constructing a logical database schema.
- Is a “blueprint” for how to store massive amounts of information in a database.
- The schema is an abstract structure that represents the logical view of the database.
- Database schema design organizes the data into separate entities, determines how to create relationships between organized entities, and how to apply the constraints on the data.
- Designers create database schemas to give other database users, such as programmers and analysts, a logical understanding of the data.

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The goals of database schema design

- Reducing or eliminating data redundancy.
- Preventing data inconsistencies and inaccuracies.
- Ensuring the correctness and integrity of data.
- Facilitating rapid data lookup, retrieval, and analysis.
- Keeping sensitive and confidential data secure, yet accessible to those who need it.

A database logical schema usually lasts for decades!

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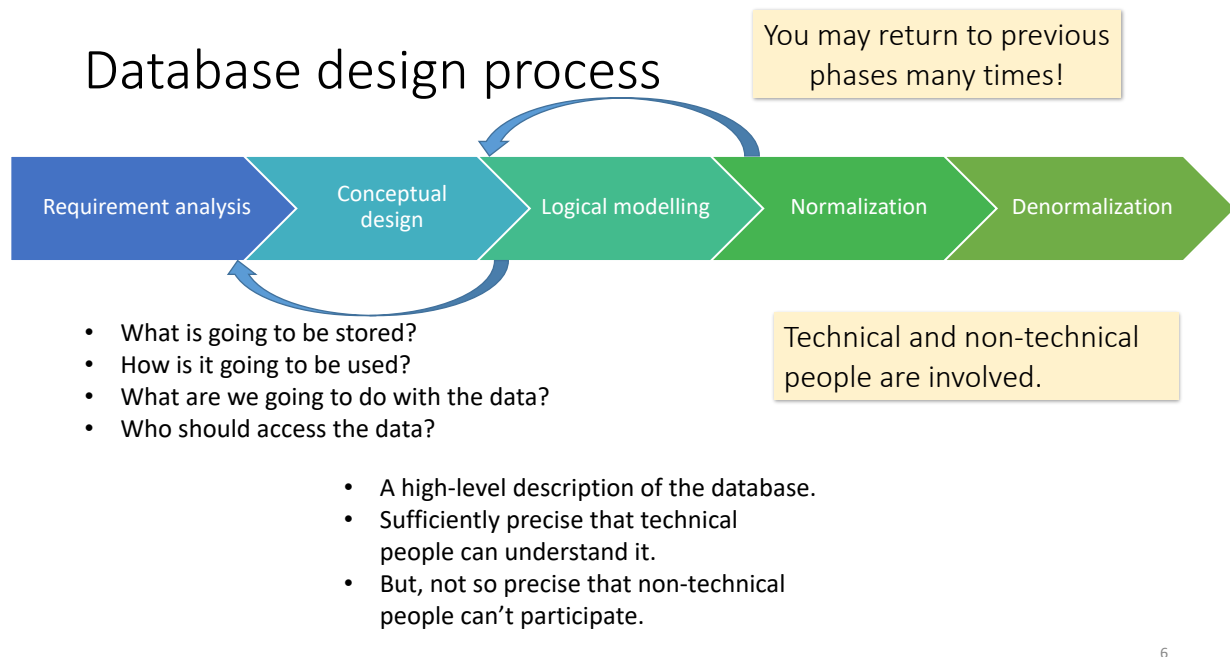
Best practices in database schema design

- **Expertise:** Understanding your data and the attributes of each element helps build out the most effective database schema design. A well-designed schema can enable data to grow exponentially.
- **Documentation:** Database schemas are useful long after they've been created, and will be viewed by many other people, which makes good documentation essential.
- **Naming conventions:** Define and use appropriate naming conventions to make your database schema designs most effective.
- **Security:** Data security starts with a good database schema design.

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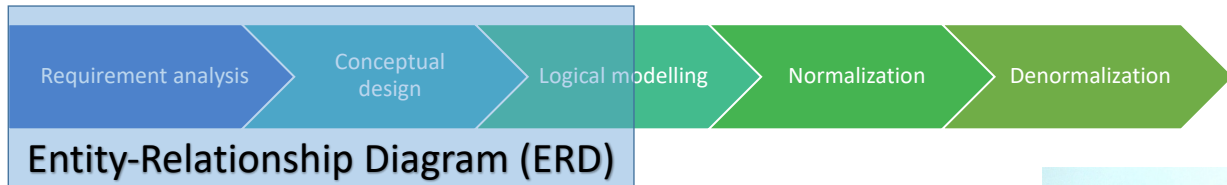
Database design process



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Entity-Relationship Diagram (ERD)



- Developed by Peter Chen, at the time at MIT
- Chen, P. P. S. (1976). The entity-relationship model—toward a unified view of data. *ACM transactions on database systems*, 1(1), 9-36.



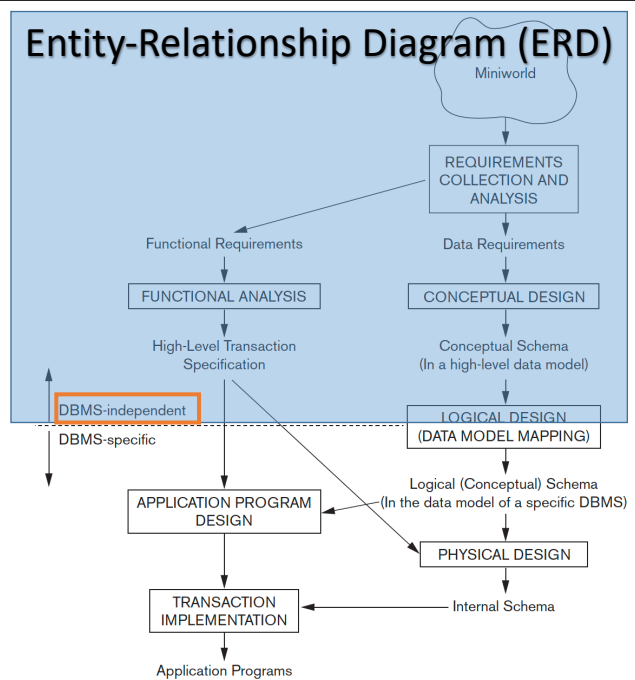
One of the most cited paper in computer science, cited more than 12000 times.

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Database design

EDR is a DBMS independent process.

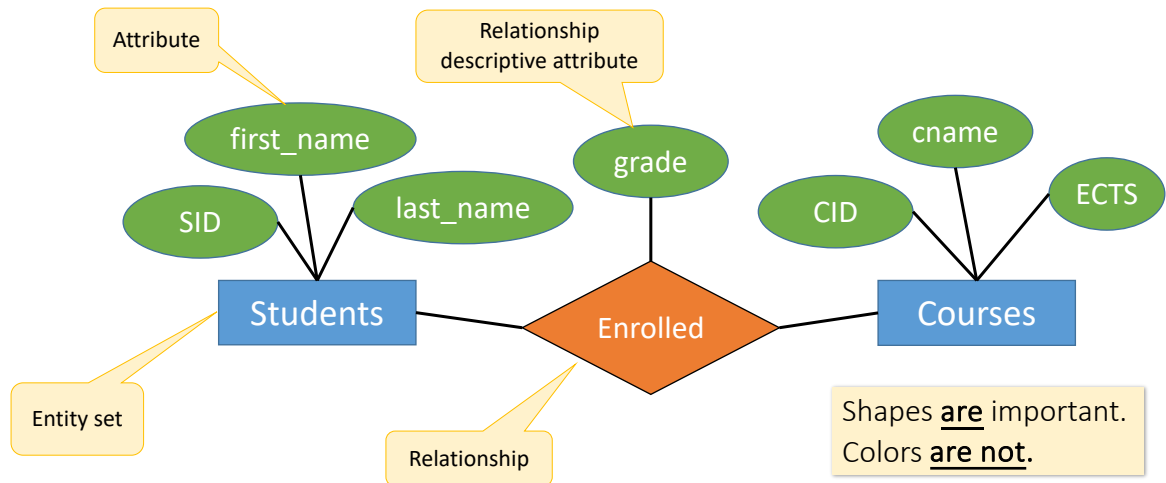


source: Elmasri, R., & Navathe, S. (2014). *Fundamentals of database systems* (Vol. 7). Pearson.

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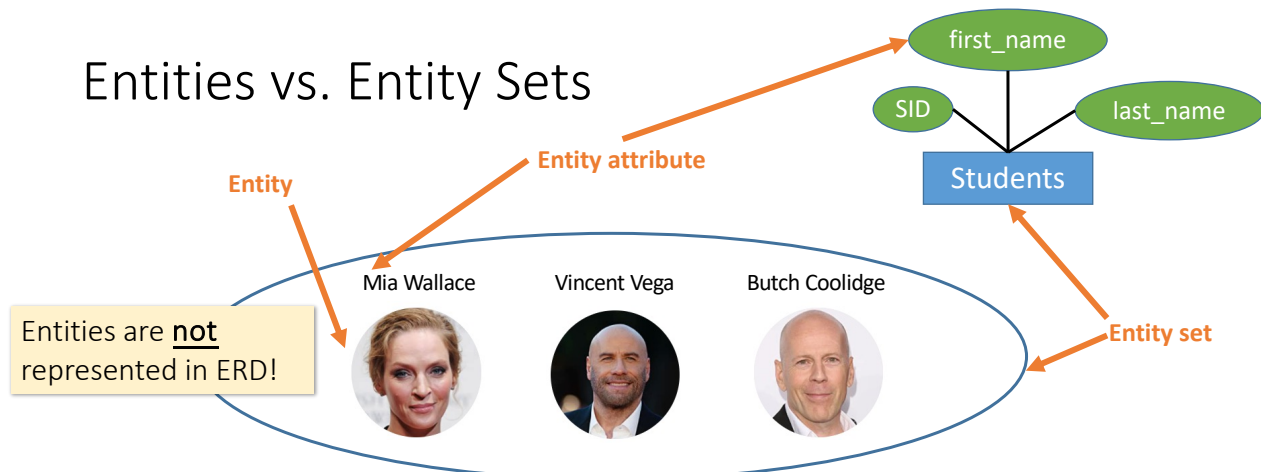
ERD – Chen notation



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Entities vs. Entity Sets



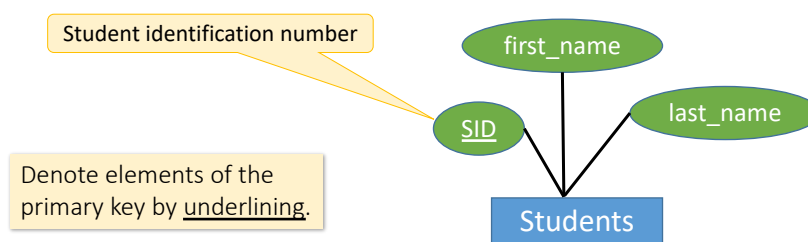
- **Entities** are the individual objects, which are members of entity sets. An entity is an object in the real world that is distinguishable from other objects.
- **Entity sets** are the *classes* or *types* of objects in our model, collection of similar entities.

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Key attribute

- A key is a **minimal** set of attributes that uniquely identifies an entity.
- There could be more than one candidate key; if so, we designate one of them as the primary key.



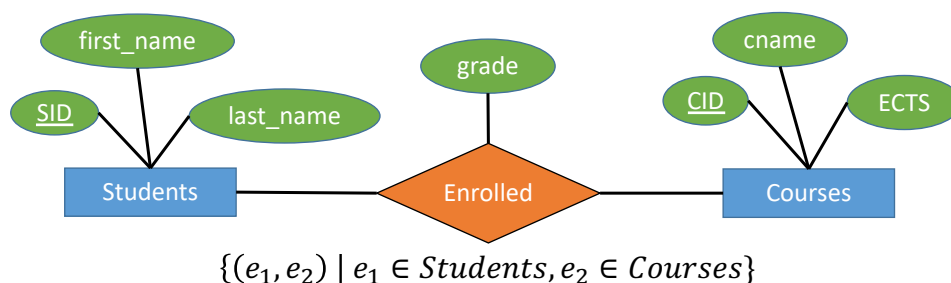
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Relationship

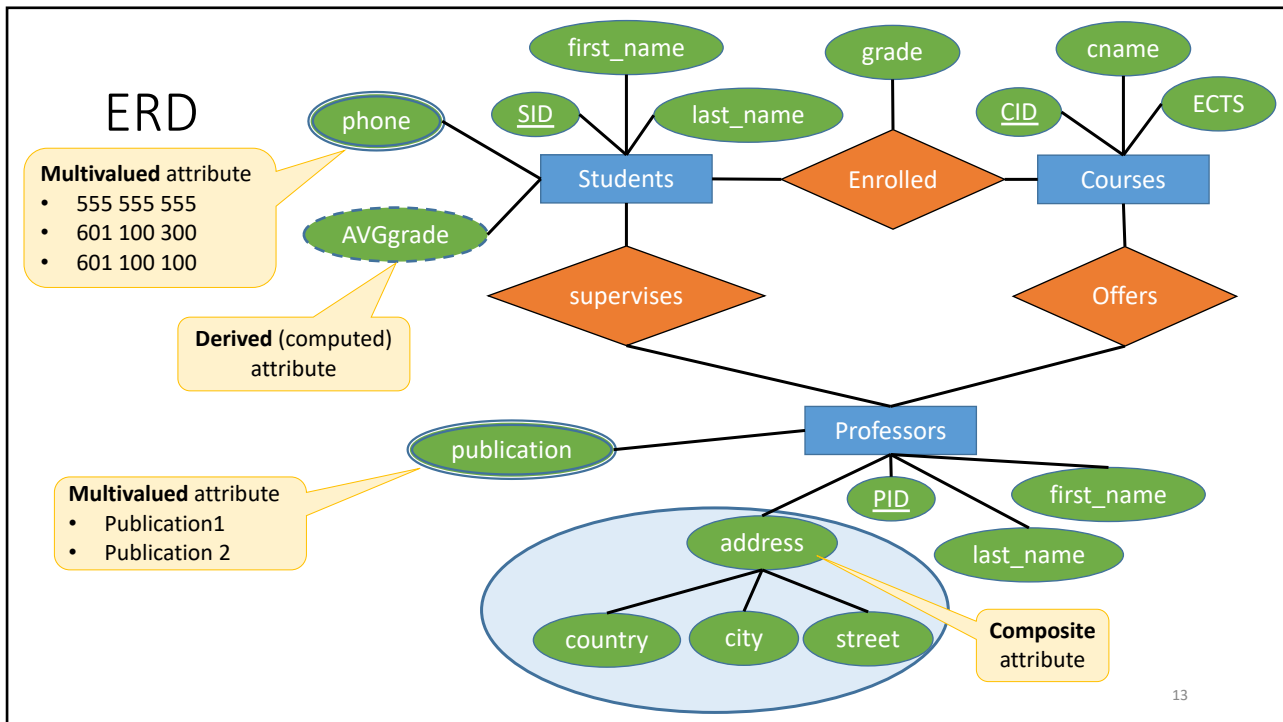
- A relationship is an association among two or more entities.
- A relationship set can be thought of as a set of n-tuples:

$$\{(e_1, e_2, \dots, e_n) \mid e_1 \in E_1, e_2 \in E_2, \dots, e_n \in E_n\}$$

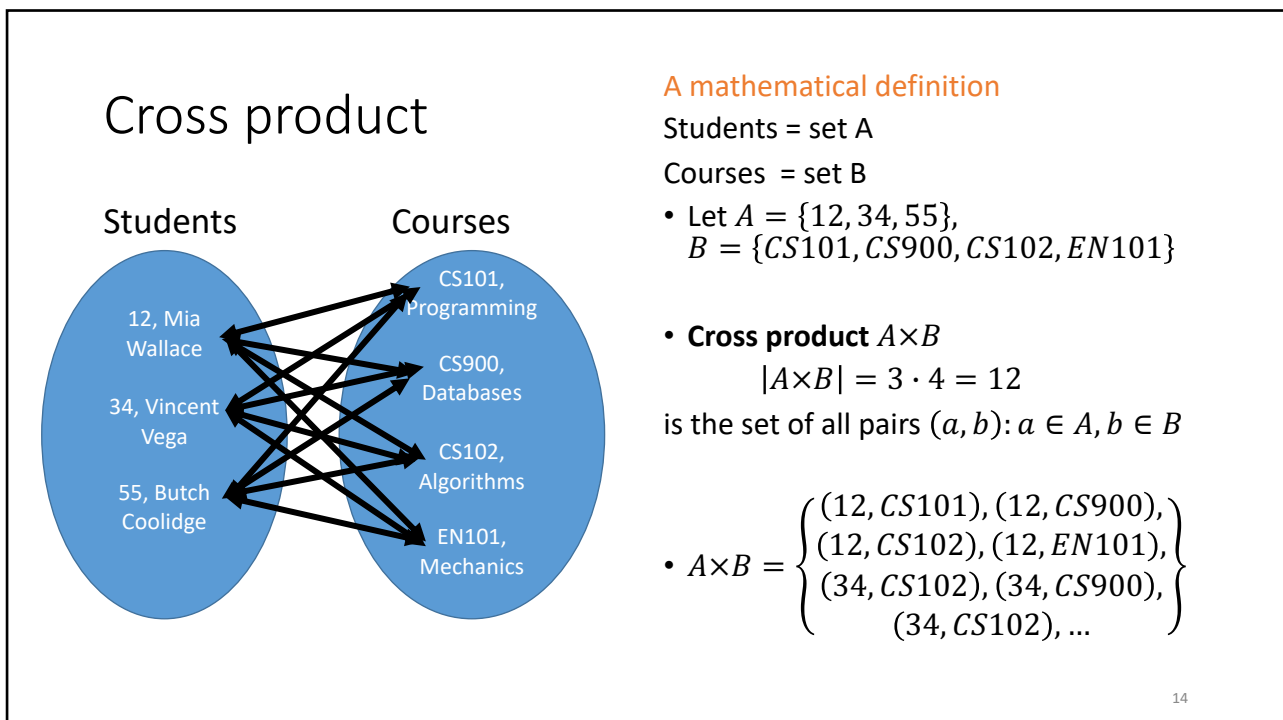


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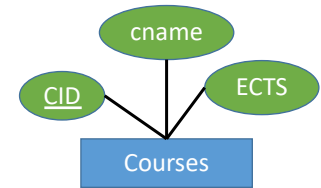
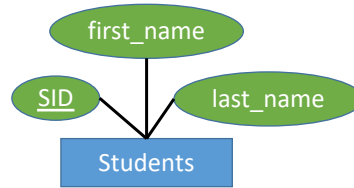


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Cross Product

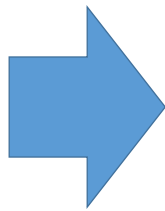


Students

sid	first_name	last_name
12	Mia	Wallace
34	Vincent	Vega
55	Butch	Coolidge

Courses

cid	cname	ECTS
CS101	Programming	5
CS900	Databases	6
CS102	Algorithms	4
EN101	Mechanics	5

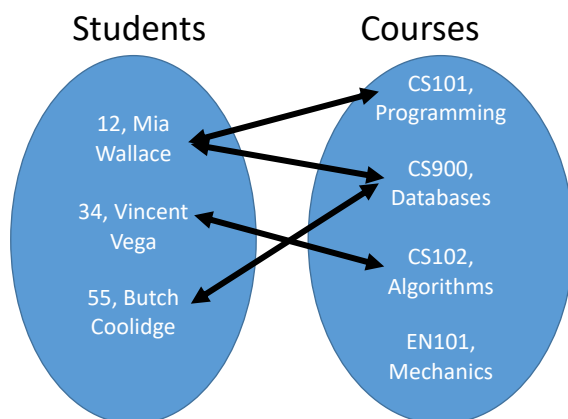


Students × Courses

sid	first_name	last_name	cid	cname	ECTS
12	Mia	Wallace	CS101	Programming	5
12	Mia	Wallace	CS900	Databases	6
12	Mia	Wallace	CS102	Algorithms	4
12	Mia	Wallace	EN101	Mechanics	5
34	Vincent	Vega	CS101	Programming	5
34	Vincent	Vega	CS900	Databases	6
34	Vincent	Vega	CS102	Algorithms	4
34	Vincent	Vega	EN101	Mechanics	5
55	Butch	Coolidge	CS101	Programming	5
55	Butch	Coolidge	CS900	Databases	6
55	Butch	Coolidge	CS102	Algorithms	4
55	Butch	Coolidge	EN101	Mechanics	5

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Relationship



A mathematical definition

Students = set A

Courses = set B

- Let $A = \{12, 34, 55\}$,
 $B = \{CS101, CS900, CS102, EN101\}$

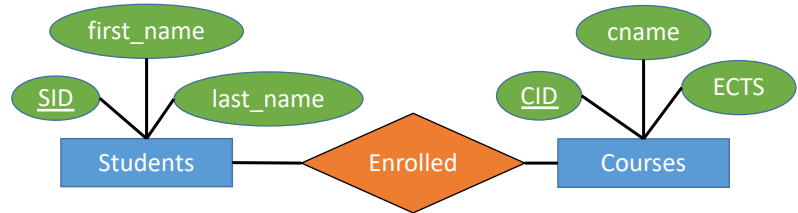
- A **relation** is a subset of $A \times B$

- $R = \{(12, CS101), (12, CS900), (34, CS102), (55, CS900)\}$

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Relationship



Students

sid	first_name	last_name
12	Mia	Wallace
34	Vincent	Vega
55	Butch	Coolidge

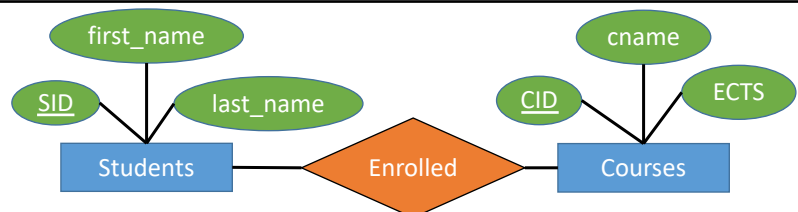
Courses

cid	cname	ECTS
CS101	Programming	5
CS900	Databases	6
CS102	Algorithms	4
EN101	Mechanics	5

- Relation: $Students \bowtie Courses$
- A **relationship** between **entity sets** *Students* and *Courses* is a **subset of all possible pairs of entities in Students and Courses**, with tuples uniquely identified by *Students and Courses's keys (SID and CID)*.

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Relationship



Students

sid	first_name	last_name
12	Mia	Wallace
34	Vincent	Vega
55	Butch	Coolidge

Courses

cid	cname	ECTS
CS101	Programming	5
CS900	Databases	6
CS102	Algorithms	4
EN101	Mechanics	5

Enrolled ($Students \bowtie Courses$)

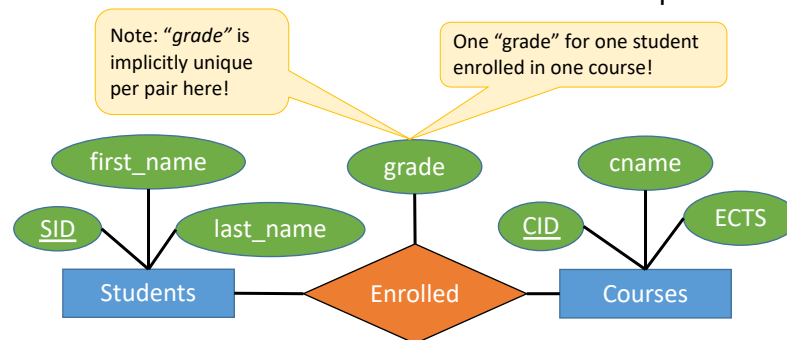
sid	first_name	last_name	cid	cname	ECTS
12	Mia	Wallace	CS101	Programming	5
12	Mia	Wallace	CS900	Databases	6
34	Vincent	Vega	CS102	Algorithms	4
55	Butch	Coolidge	CS900	Databases	6

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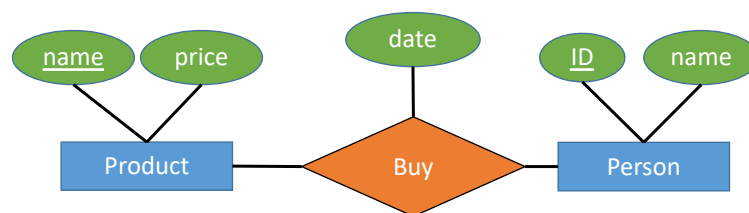
Relationship and attributes

- There can only be **one relationship** for every **unique combination of entities**.
- This also means that **the relationship is uniquely determined by the keys of its entities**.
- This follows from our mathematical definition of a relationship – it's a SET!



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Relationship vs. Entity?



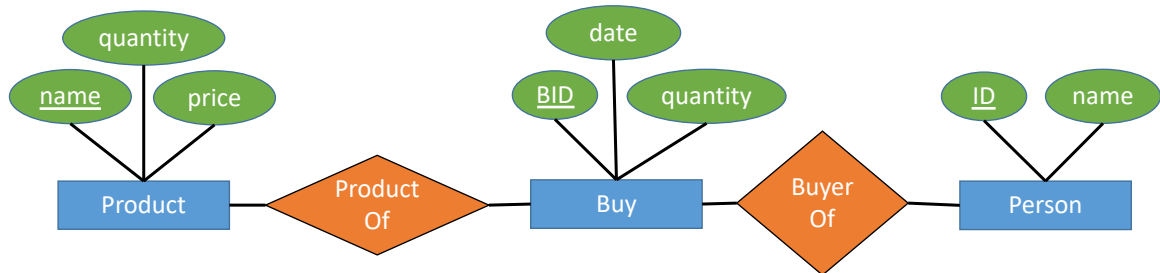
- A person can only buy a specific product once

Modeling something as a relationship makes it unique

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Relationship vs. Entity?



- Now we can have multiple purchases per product, person pair!

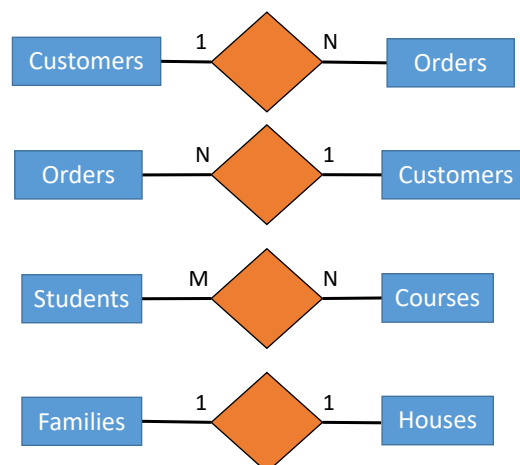
We can always use **a new entity** instead of a relationship.
For example, to permit multiple instances of each entity combination!

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Cardinality

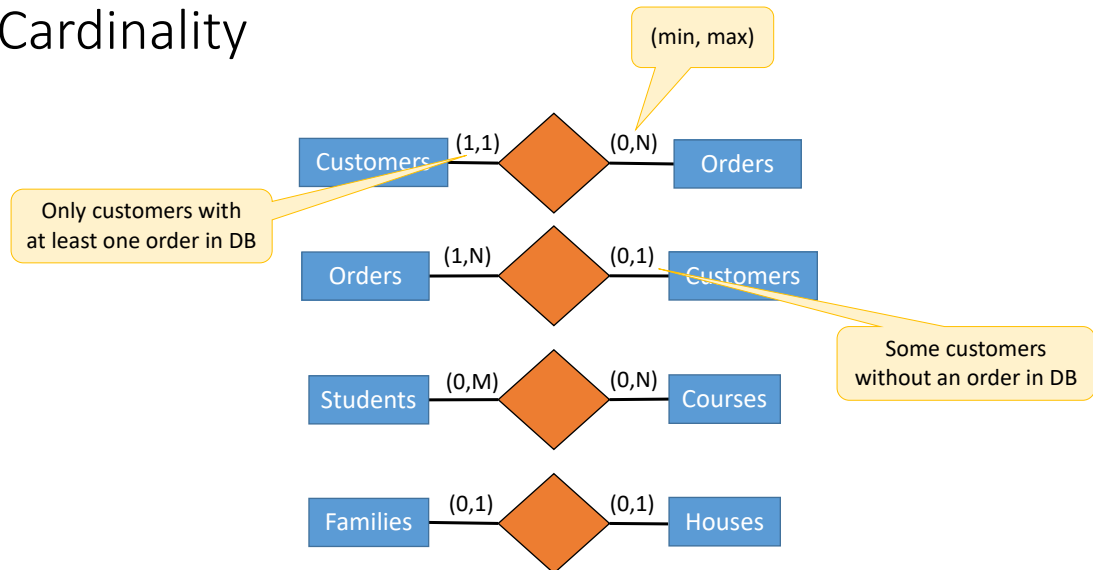
- Each customer can have many orders.
One order has one customer.
- Many students can be enrolled in many courses.
- One family lives in one house.



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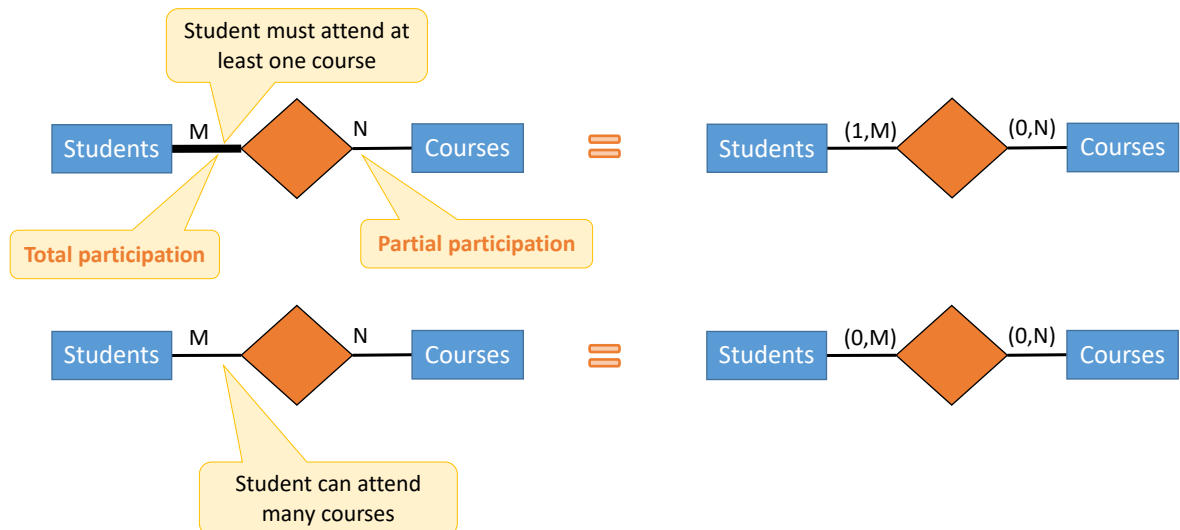
Cardinality



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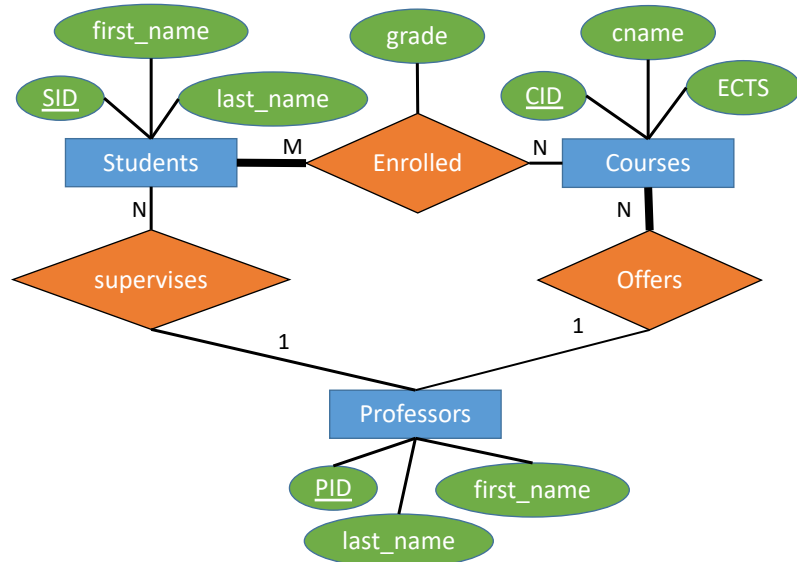
Participation



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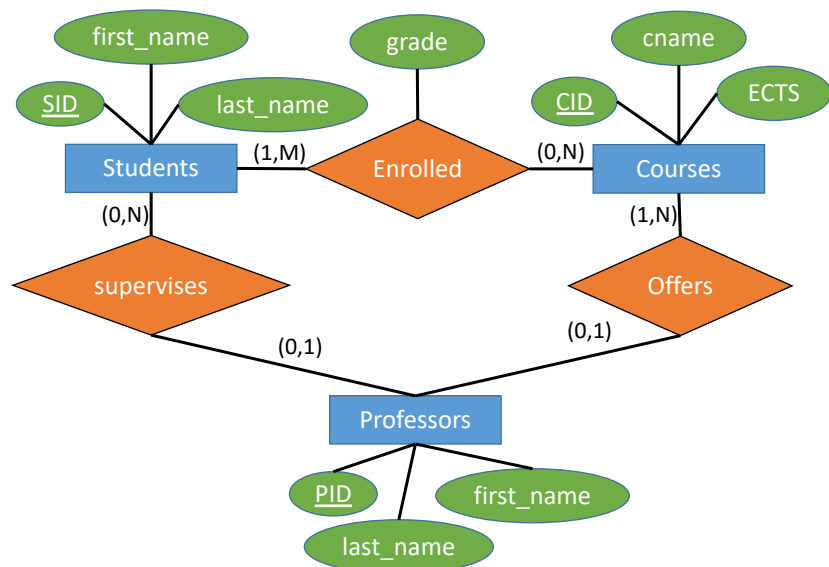
ERD



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ERD

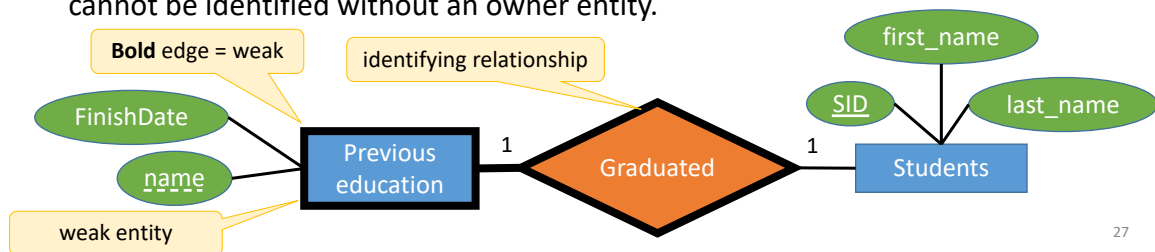


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Weak entities and identifying relationships

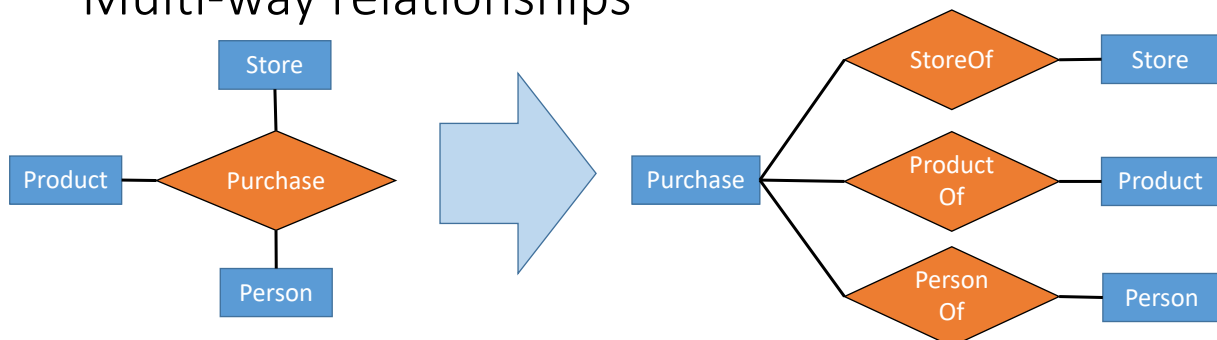
- Weak entities do **not have key attributes** of their own.
- Weak entity is identified by being related to **identifying (owner) entity** in combination with one of their attribute values.
- Identifying relationship** relates a weak entity type to its owner.
- A weak entity type always has a **total participation constraint** (existence dependency) with respect to its identifying relationship because a weak entity cannot be identified without an owner entity.



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Multi-way relationships

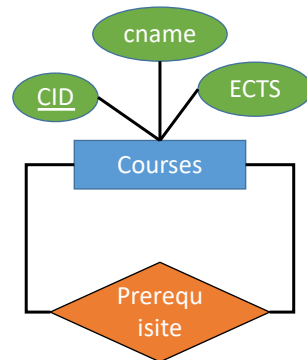


- Binary relationships are useful if we want to have multiple instances of the “relationship” per entity combination.
- It is almost always possible to convert a multilateral relationship into a bilateral one.
- Some database design tools permit only binary relationships.

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Recursive (unary) relationships



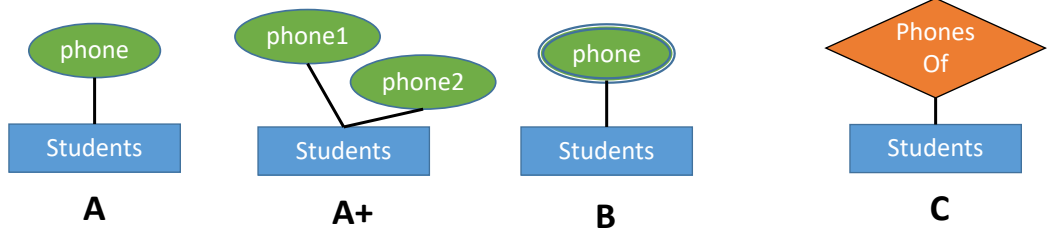
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Entity vs. attribute

Requirement: Student has a phone number.

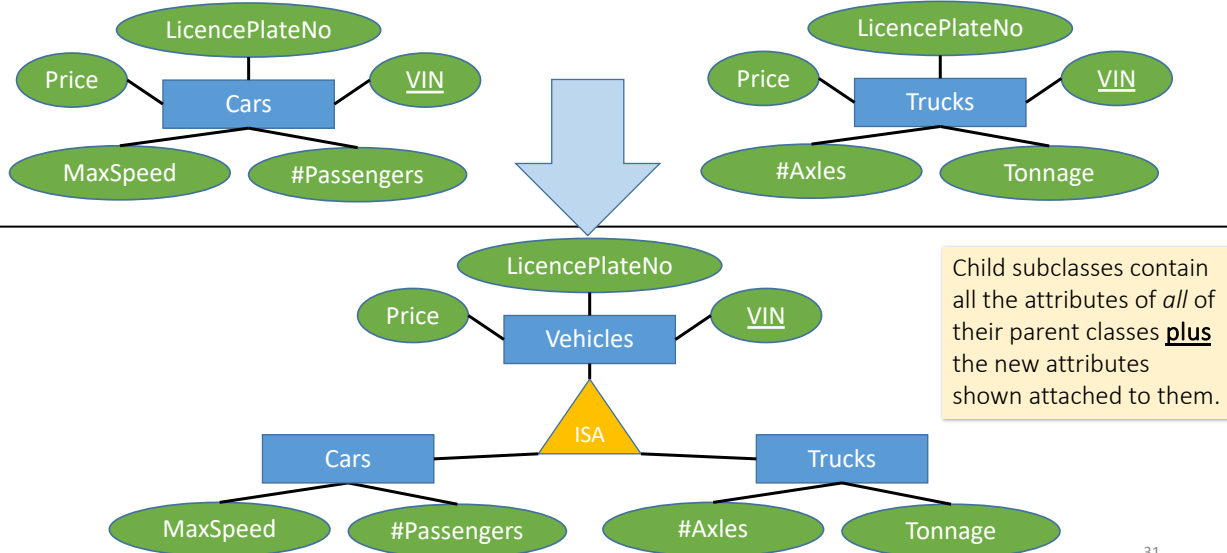
- **A** – one phone number per student
- **B** – possible many phone numbers per student
- **C** – possible many phone numbers per student + additional information



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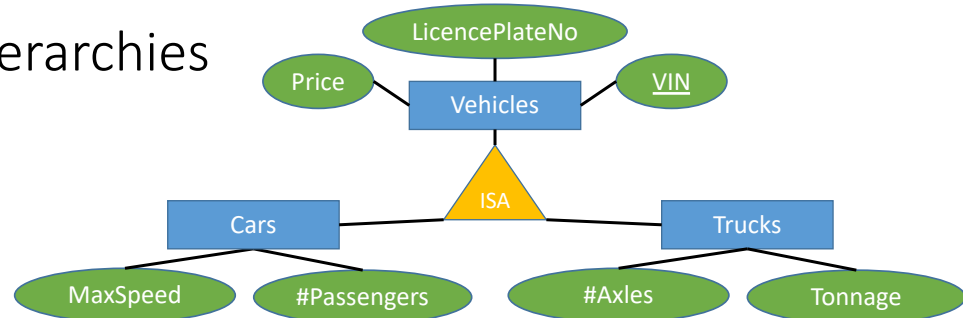
Class hierarchies



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Class hierarchies

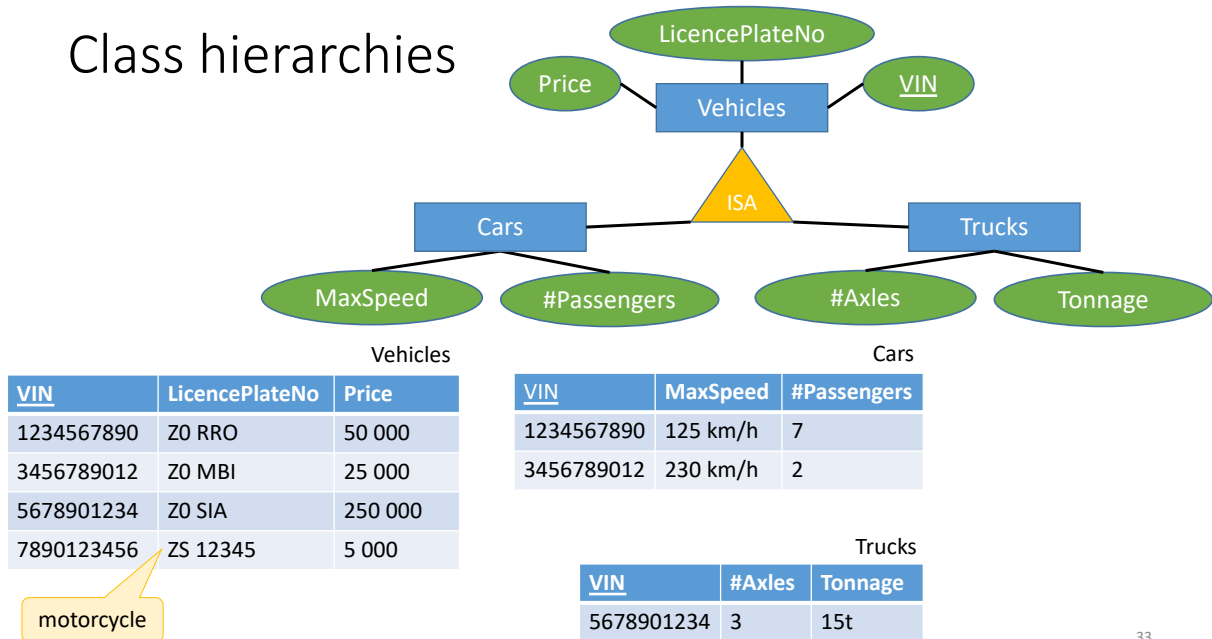


- **Overlap** constraints determine whether two subclasses are allowed to contain the same entity. By default, entity sets are constrained to have no overlap. There is no vehicle which is both a car and a truck.
- **Covering** constraints determine whether the entities in the subclasses collectively include all entities in the superclass. By default, there is no covering constraint. We can have vehicles that are not cars or trucks.

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Class hierarchies

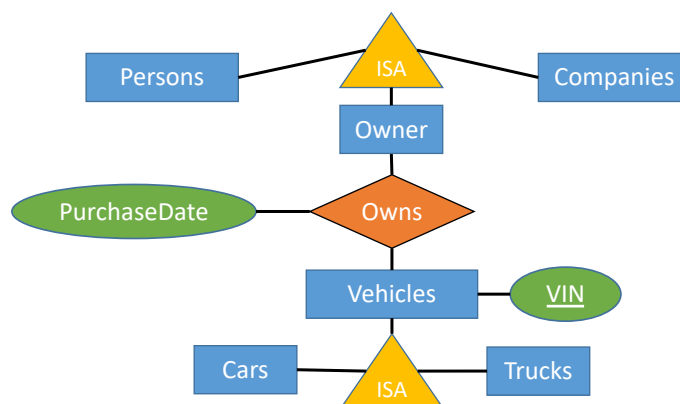


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Union types with class hierarchies

- A vehicle can be owned either by a person or by a company, but not by both.
- How to model it?

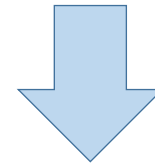
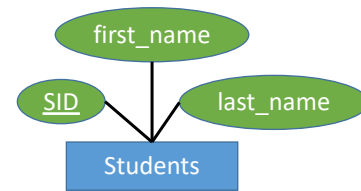


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ERD to logical schema

- An entity set becomes a relation (table)



Students

<u>SID</u>	first_name	last_name

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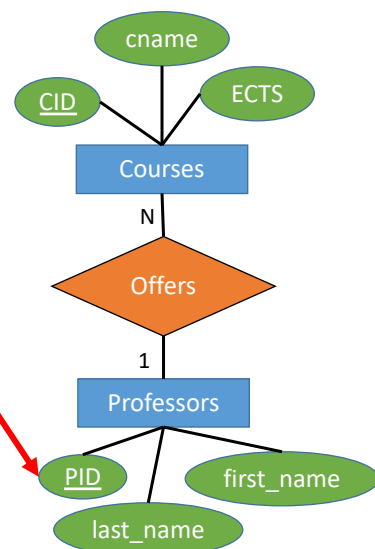
ERD to logical schema 1:N

Courses

<u>CID</u>	cname	ECTS	PID

Professors

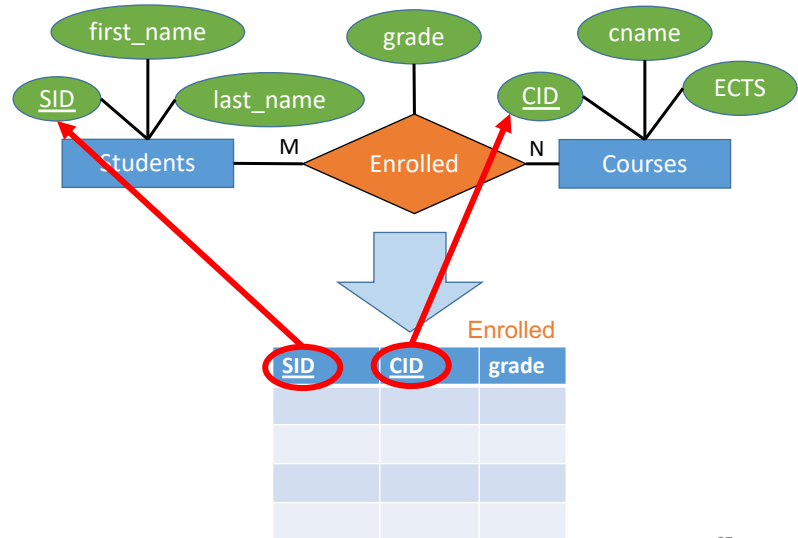
<u>PID</u>	first_name	last_name



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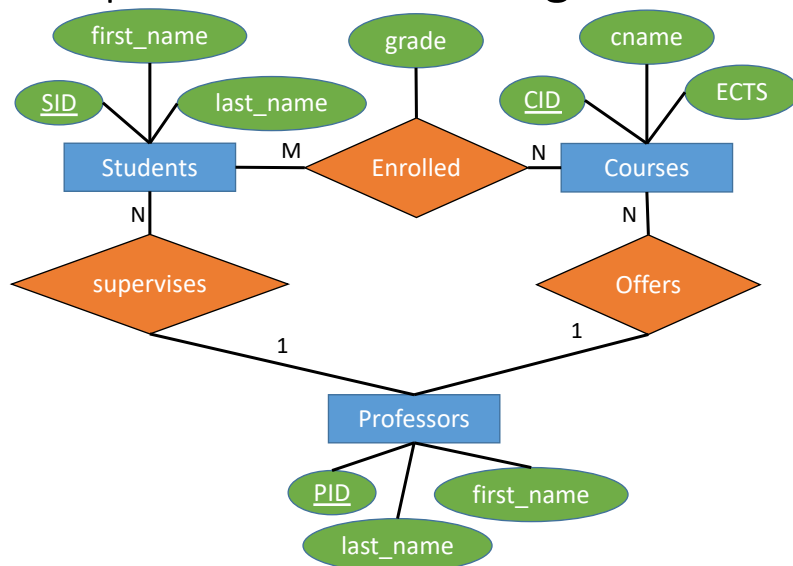
ERD to logical schema M:N



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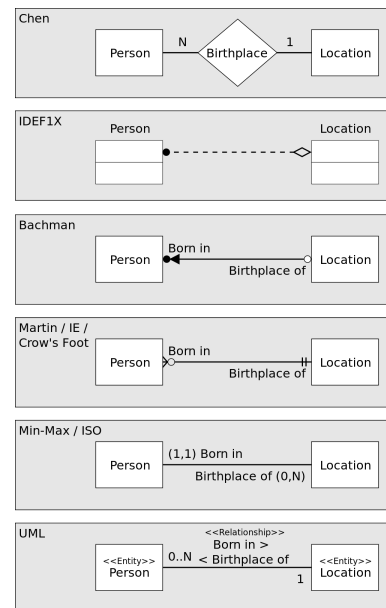
How to represent this as a logical schema?



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Other notations

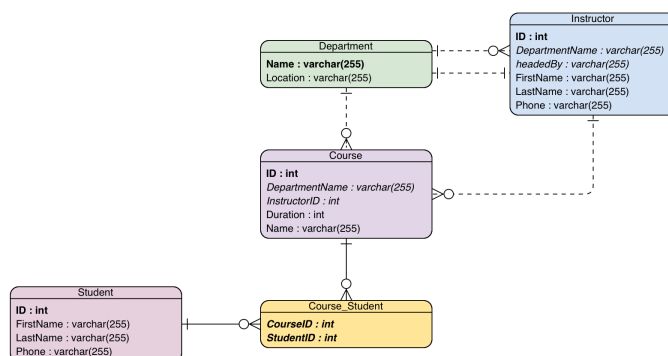


source: Wikipedia

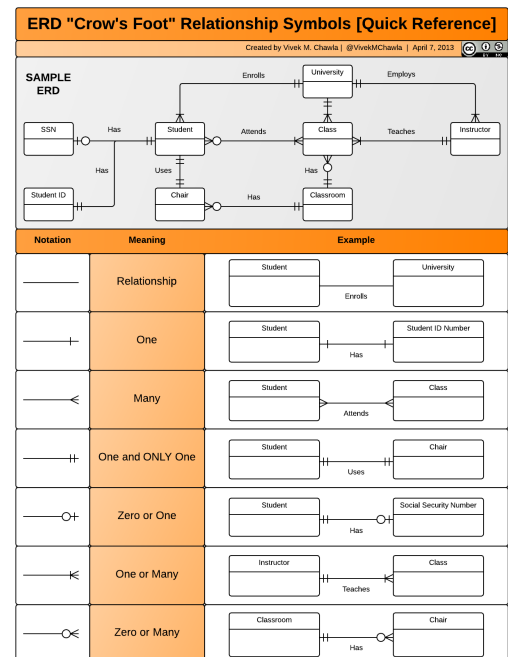
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Other notations Crown's Foot

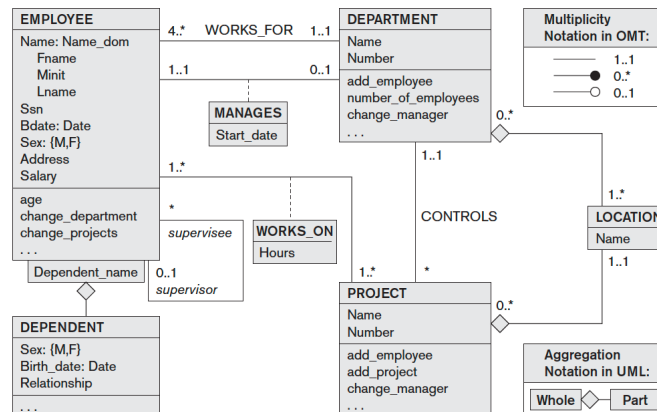


source: visual-paradigm.com



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Other notations UML class diagram



source: Elmasri, R., & Navathe, S. (2014). *Fundamentals of database systems* (Vol. 7). Pearson.

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Thank you for your attention!

Time for questions

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